



Nguyen Khanh Toan

SOC Analyst Intern

Phone: +84 77 266 0810
Email: tn421015@gmail.com
LinkedIn: linkedin.com/in/tofan0810
GitHub: github.com/tofan0810
Portfolio: tofan0810.github.io/my-portfolio/
Address: Ho Chi Minh City, Vietnam

Cybersecurity student specializing in Blue Team & SOC operations — hands-on with Wazuh SIEM/XDR deployment, Sysmon log triaging, custom detection engineering (MITRE ATT&CK-mapped), and active response automation. Certified SOC Analyst (LetsDefend) proficient in alert investigation, threat containment, and phishing triage.

Cybersecurity Projects

Centralized Wazuh SIEM & Detection Engineering Lab

[GitHub](#)

Independent Project

- **Core Tech:** Wazuh SIEM/XDR (Indexer, Manager, Dashboard), Docker Compose, Windows Sysmon v15.2, Kali Linux, Apache/XAMPP, MITRE ATT&CK
- Deployed a full-stack Wazuh SIEM/XDR on Ubuntu Server via Docker Compose (Single-Node); configured Wazuh Agent + custom Sysmon XML rules on Windows 10 to collect kernel-level endpoint telemetry (Windows Event Logs, process creation, network connections) for **centralized security monitoring**.
- Authored Custom XML Decoders (Regex-based URL parser for Apache access logs) and Custom Detection Rules mapped to **MITRE ATT&CK**: Rule 100001 (Level 12, T1110 – RDP Brute Force, frequency: 5 failures/60s) and Rule 100002 (Level 10, T1190 – LFI/Directory Traversal, PCRE2 Regex covering single & double URL encoding); validated rule logic via `wazuh -logtest` to **reduce false positives and improve alert fidelity**.
- Configured **Active Response** automation: on alert trigger, Wazuh Manager remotely executed `netsh.exe` on the Windows Agent to auto-block attacker IPs via Windows Defender Firewall for 600s — demonstrating an end-to-end **Detect → Alert → Contain** SOAR-like workflow with zero manual intervention.

Secure FTP Client with Real-Time Virus Scanning

[GitHub](#)

Computer Networks Lab

- Core Tech: Python, Socket Programming, ClamAV/ClamD, FTP Protocol, FileZilla Server
- Developed a multi-component secure file transfer infrastructure with real-time **malware analysis** capabilities to address vulnerabilities in legacy FTP transfers.
- Built an interactive Python FTP Client (PASV mode, multi-file transfers) coupled with a `clamav_agent.py` listener for **threat detection**; configured ClamD to return **malware signatures** dynamically; enforced network rules via Windows Firewall.
- Guaranteed 100% inspection of incoming files before storage, effectively preventing EICAR test malware from infecting server paths through **behavioral analysis** and signature-based detection.

TravelTogether – Smart Tourism Platform (Security Module)

[GitHub](#)

Collaborative Project

- **Core Tech:** FastAPI, PostgreSQL (Supabase cloud), Firebase Admin SDK, JWT, AES Encryption, Flutter
- Architected the security module for a cloud-backed platform, implementing authentication workflows and an automated emergency safety system — interfacing directly with **cloud-based APIs** (Firebase Admin SDK, Supabase REST) to enforce access control and protect user privacy.
- Enforced JWT-based session validation via Firebase Admin SDK; engineered an autonomous Dual-PIN security module that monitors user safety timestamps and auto-triggers encrypted emergency GPS alerts via email on 36-hour inactivity — end-to-end **scripting automation** against cloud service APIs.
- Eliminated identity-spoofing attack vectors and established a privacy-first backend architecture; conducted **security testing** and vulnerability assessment to harden API endpoints against authentication bypass.

Threat Investigation & CTF Repository [Blog](#)

Continuous Practice

- Investigated threat artifacts across security challenges; extracted IoCs, analyzed PCAPs (Wireshark), and mapped TTPs to MITRE ATT&CK.
- Conducted memory forensics and binary analysis via GDB/Ghidra; built Python scripts to parse hex dumps and isolate payloads.
- Maintained detailed investigation runbooks and writeups adhering to enterprise SOC documentation standards.

Technical Skills

- **SOC Operations & Incident Handling:** Alert Triaging, Log Correlation, Root-Cause Analysis, False-Positive Reduction, Phishing & Malware Triage, NIST Incident Response Lifecycle
- **SIEM & Detection Engineering:** Wazuh SIEM/XDR (rule authoring, XML decoders, alert tuning), `wazuh-logtest`, Alert Fidelity Optimization, Active Response (SOAR automation)
- **Endpoint & Network Defense:** Windows Sysmon Telemetry (process creation, network events), Wireshark, PCAP Analysis, Firewall Enforcement (netsh, iptables)
- **Security Frameworks:** MITRE ATT&CK (Enterprise TTPs, T1110, T1190), OWASP Top 10, Cyber Kill Chain
- **Programming & Automation:** Python (log parsing, forensics scripts, security automation), Bash Scripting, C/C++, SQL
- **Systems & Hardening:** Kali Linux, Ubuntu Server, Windows 10/Server, Docker, Cloud & REST API Security

Certifications

SOC Analyst Learning Path

LetsDefend • Verify

Sep 2026

- Hands-on practical training covering Log Analysis, Incident Handling, Threat Detection, Phishing Analysis, and Endpoint/Network Investigation in simulated enterprise environments.

Education

VNU-HCM University of Science (HCMUS) B.Sc. Information Technology, Major in Cybersecurity

2024 – Present

- **GPA: 3.5/4.0** • Ho Chi Minh City, Vietnam
- Flexible schedule; fully committed and available for 3–6 months on-site internship
- Relevant Coursework: Computer Networks, Operating Systems, Network Security Basics, Systems Architecture

Nguyen Dinh Chieu High School for the Gifted Major in Mathematics

Graduated Jun 2024

- 2nd Prize in Provincial Physics Competition; Dong Thap, Vietnam

Languages

English	Advanced
Vietnamese	Native